

From Bits to Qubits: The Growth Story of Quantum Computing

Quantum computing may still be in the early stages of evolution, but its potential impact on everyday life is significant. We delve into the key concepts behind it, the reasons for its rapid growth, and how global advancements are shaping its future.



Imagine you're searching for a specific book on Amazon, a platform with millions of titles. Now, let's assume you're using a powerful computer with an Intel i7 processor running at around 5GHz. In this scenario, the i7 processor isn't just running the Amazon website on your device, but also assumed to be searching through Amazon's massive database of millions of books. While the Intel i7 is fast for many tasks, searching such a large database with just this processor would still take considerable time. Even with multiple cores, each core can handle only one part of the search at a time, which means the process could take several

minutes, especially when checking millions of records. Today, however, systems like Amazon's can return results in seconds, thanks to advanced algorithms, distributed systems, and parallel processing across many servers—not just the power of a single i7 processor.

Let's explore another amazing feature beyond individual computing: distributed computing. This technique drives most modern computing systems. In distributed computing, multiple processors or cores in a network of computers work together to solve a problem faster. For example, Amazon's search engine leverages this technology when you search for a

product. Behind the scenes, Amazon's system operates on a distributed network of hundreds of powerful servers, each equipped with multiple processors. When you search for an item, the search request is split into many parts, and different servers handle different sections of the search simultaneously, dramatically speeding up the process. This allows Amazon to quickly scan through millions of product listings and return relevant results in just a few seconds. However, even with such advanced distributed computing, each server and processor still works step by step through its assigned data. While multiple servers may handle different sections of the